American Computer Science League

2021-2022 ● Contest 3: Short Problems ● Intermediate Division

1. Boolean Algebra

Simplify the following Boolean expression:

$$\overline{A\,\overline{B}} \;+\; \overline{\overline{A}\,B\,\overline{C}} \;+\; \overline{\overline{B}\,C}$$

A. 0
B. 1
C. B
D. $\overline{A} + \overline{B} + \overline{C}$
E. $\overline{A} + \overline{B} + \overline{C}$

2. Boolean Algebra

How many ordered triple(s) make the following Boolean expression TRUE?

$$A\,\overline{B} \;+\; \overline{A}\,(B \;+\; \overline{C}\,)$$

A. 0
B. 3
C. 5
D. 7
E. 8

3. Data Structures

How many nodes have only a right child in the binary search tree for:

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A. 0
B. 1
C. 2
D. 3
E. 4

4. Data Structures

Given an initially empty stack and the following commands on the stack, what will be the next item popped?

PUSH("C"), PUSH("A"), PUSH("N"), X=POP( ),
X=POP( ), PUSH("I"), X=POP( ), PUSH("S"),
PUSH("M"), X=POP( ), X=POP( ), PUSH("A"),
PUSH("J"), X=POP( ), PUSH("O"), X=POP( ),
PUSH("R"), X=POP( )

A. S
B. M
C. I
D. C
E. A

5. FSAs and Regular Expressions

A new company is allowing its employees to select their own phone number and extension if it satisfies this regular expression:

1-321-[^05][0-9][^5]-[12][3-6][^7-9]8-[^0][2-6]

Which of the following are acceptable phone numbers and extensions?

a. 1-321-123-2348-11
b. 1-321-155-2368-96
c. 1-321-444-2448-44
d. 1-321-927-2408-84
e. 1-321-559-2328-13
f. 1-321-306-1366-62

A. a, f
B. b, c
C. e, f
D. c, d
E. a, e

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1. 布尔代数

化简**以下布尔表达式：**

$$\overline{A\,\overline{B}} + \overline{\overline{A}\,B\,\overline{C}} + \overline{\overline{B}\,C}$$

A. 0
B. 1
C. B
D. $\overline{A} + \overline{B} + \overline{C}$
E. $\overline{A} + \overline{B} + \overline{C}$

2. 布尔代数

有多少个**有序三元组可以使**得下述**布尔表达式结果为**真？

$$A\,\overline{B} + \overline{A}\,(B + \overline{C})$$

A. 0
B. 3
C. 5
D. 7
E. 8

3. 数据结构

在下述二叉搜索树中，有多少个结点有且仅有一个右孩子结点？

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A. 0
B. 1
C. 2
D. 3
E. 4

4. 数据结构

给定一个初始为空的栈，并在栈上执行以下命令。**请问下一个**将要出栈的元素是什么?

PUSH(“C”), PUSH(“A”), PUSH(“N”), X=POP( ),
X=POP( ), PUSH(“I”), X=POP( ), PUSH(“S”),
PUSH(“M”), X=POP( ), X=POP( ), PUSH(“A”),
PUSH(“J”), X=POP( ), PUSH(“O”), X=POP( ),
PUSH(“R”), X=POP( )

A. S
B. M
C. I
D. C
E. A

5. FSAs 和正则表达式

一家新开的公司允许员工选择自己的电话号码和分机号，只要**号码**满足**以下**正则表达式：

1-321-[^05][0-9][^5]-[12][3-6][^7-9]8-[^0][2-6]

以下哪些电话号码和分机号**满足条件**?

a. 1-321-123-2348-11
b. 1-321-155-2368-96
c. 1-321-444-2448-44
d. 1-321-927-2408-84
e. 1-321-559-2328-13
f. 1-321-306-1366-62

A. a, f
B. b, c
C. e, f
D. c, d
E. a, e